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System and method for discovery and binding of remote access controller firmware to a proxy service

Abstract

In one or more embodiments, one or more systems may comprise a Remote Access Controller (RAC) configured with a minimum number of built-in features and a private cloud server (PCS) communicatively coupled to the RAC and storing a set of available services retrieved from an external cloud server. The RAC may be configured to communicate with a DHCP service, an onboarding service and a RAC proxy service to send a RAC hardware identity certificate and receive a RAC proxy service certificate. If the RAC hardware identity certificate and the RAC proxy service certificate are authenticated, the RAC proxy service may establish a binding with RAC firmware.

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Background/Summary

BACKGROUND
Field of the Disclosure
(1) This disclosure relates generally to information handling systems and more particularly to remote access controllers (RACs) in servers in data centers, and more particularly to systems and methods for monitoring and managing RACs.
Description of the Related Art
(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling

needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

SUMMARY

(3) Embodiments may be directed to system on a server comprising a plurality of devices. The system comprises a remote access controller (RAC) and a private cloud server (PCS) communicatively coupled to the server and an external cloud server. The RAC comprises a RAC processor and a memory storing: a set of controls, the first set of controls comprising a power control for monitoring power supplied to the server and a thermal control for monitoring a temperature of the server; a hardware abstraction layer (HAL) for communicating with the plurality of devices; a Desktop Bus (D-Bus) communications bus for communicating with the HAL and facilitate communications between a set of applications executing on the server; a D-Bus to Remote Procedure Call (RPC) mapper for translating communications from one or more devices of the plurality of devices into an RPC format; a D-Bus to Representational State Transfer (REST) mapper for translating communications from one or more devices of the plurality of devices into a REST Application Programming Interface (API) format; a Remote Procedure Call (RPC) queue to facilitate communication between two or more applications of the set of applications executing on the server; and a set of instructions executable by the RAC processor to execute the controls stored in the memory in the RAC. The PCS comprises: a PCS processor and a memory storing a set of instructions executable by the PCS processor for: communicating with the external cloud server to retrieve a set of available services from a plurality of available services; and executing the set of available services for processing information on the server.

(4) In some embodiments, the set of PCS instructions are executable by the PCS processor for: determining an available service of the set of available services stored in the memory in the PCS is needed by the server to process a set of information; installing the available service in a set of services in the memory in the RAC; determining when the available service has processed the information; and uninstalling the available service from the set of services in the memory in the RAC.

(5) In some embodiments, the set of PCS instructions are executable by the PCS processor for determining an available service of the set of available services stored in the memory in the PCS increases the scope of operation of the server to process the set of information. In some embodiments, the available service comprises one of a credentialing service, a telemetry service, an advanced power service or an advanced thermal service. In some embodiments, the set of services comprises a time-critical service. In some embodiments, the time-critical service comprises one or more of an advanced power service, a thermal service or a telemetry service. In some embodiments, the memory in the PCS stores a unified database storing information for one or more of the controls and one or more of the available services in the set of available services; and a management service orchestrator (MSO) for resolving conflicts between two or more of the available services in the set of available services executing on the PCS.

(6) Embodiments may be directed to a data center comprising plurality of servers. Each server comprises a remote access controller (RAC) comprising a RAC processor and a memory. The memory stores a set of controls, the set of controls comprising a power control for monitoring power supplied to the server and a thermal control for monitoring a temperature of the server; a hardware abstraction layer (HAL) for communicating with the plurality of devices; a Desktop Bus

(D-Bus) communications bus for communicating with the HAL and facilitate communications between a set of applications executing on the server; a D-Bus to Remote Procedure Call (RPC) mapper for translating communications from one or more devices of the plurality of devices into an RPC format; a D-Bus to Representational State Transfer (REST) mapper for translating communications from one or more devices of the plurality of devices into a REST Application Programming Interface (API) format; a Remote Procedure Call (RPC) queue to facilitate communication between two or more applications of the set of applications executing on the server; and a set of instructions executable by the RAC processor to execute the controls stored in the memory in the RAC. At least one server of the plurality of servers comprises a private cloud server (PCS) communicatively coupled to the plurality of servers and an external cloud server, the private cloud server comprising: a PCS processor; and a memory storing a set of instructions executable by the PCS processor to: communicate with the external cloud server to retrieve a set of available service from a plurality of available services; store the set of available services; and execute an available service of the set of available services to process the information on a set of servers of the plurality of servers.
(7) In some embodiments, the set of PCS instructions are executable by the PCS processor to: determine an available service of the set of available services stored in the memory in the PCS is needed by the set of servers to process the information; install the available service in the memory in the RAC in each server of the set of servers; determine when the available service has processed the information; and uninstall the available service from the memory in the RAC in each server of the set of servers. In some embodiments, the set of PCS instructions are executable by the PCS processor to determine an available service of the set of available services stored in the memory in the PCS increases the scope of operation of the set of servers of the plurality of servers to process the information. In some embodiments, the available service comprises one of a credentialing service, a telemetry service, an advanced power service or an advanced thermal service. In some embodiments, the available service comprises a time-critical service. In some embodiments, the time-critical service comprises one or more of an advanced power service, a thermal service or a telemetry service. In some embodiments, the memory in the PCS stores: a unified database storing information for one or more of the controls and one or more of the available services of the set of available services; and a management service orchestrator (MSO) for resolving conflicts between two or more of the available services of the set of available services executing on the PCS.
(8) Embodiments may be directed to a method of operating a data center comprising a plurality of servers. The method comprises storing, on each server of the plurality of servers: a set of controls, the first set of controls comprising a power control for monitoring power supplied to the server and a thermal control for monitoring a temperature of the server; a hardware abstraction layer (HAL) for communicating with a plurality of devices; a Desktop Bus (D-Bus) communications bus for communicating with the HAL and facilitate communications between a set of applications executing on the server; a D-Bus to Remote Procedure Call (RPC) mapper for translating communications from one or more devices of the plurality of devices into an RPC format; a D-Bus to Representational State Transfer (REST) mapper for translating communications from one or more devices of the plurality of devices into a REST Application Programming Interface (API) format; a Remote Procedure Call (RPC) queue to facilitate communication between two or more applications of the set of applications executing on the server; and a set of instructions executable by the RAC processor to execute the controls stored in the memory in the RAC. The method may also comprise retrieving, by a private cloud server from an external cloud server over a network, an available service of a plurality of available services; and executing the set of available services for processing information on a set of servers of the plurality of servers.
(9) In some embodiments, the method comprises determining, by the PCS processor, an available service of the set of available services stored in the memory in the PCS is needed by the set of servers to process the information; installing the available service in the memory in the RAC in

each server of the set of servers; determining when the available service has processed the information; and uninstalling the available service from the memory in the RAC in each server of the set of servers. In some embodiments, determining an available service of the set of available services stored in the memory in the PCS is needed by the set of servers to process the information comprises determining an available service of the set of available services stored in the memory in the PCS increases the scope of operation of the set of servers of the plurality of servers to process the information. In some embodiments, the available service comprises one of a credentialing service, a telemetry service, an advanced power service or an advanced thermal service. In some embodiments, the available service comprises a time-critical service. In some embodiments, the time-critical service comprises one or more of an advanced power service, a thermal service or a telemetry service. In some embodiments, the method comprises storing, in a unified database, a set of information for one or more of the controls and one or more of the available services of the set of available services; and resolving, by a management service orchestrator (MSO), conflicts between two or more of the available services of the set of available services executing on the PCS.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present disclosure and its features/advantages, reference is now made to the following description, taken in conjunction with the accompanying drawings, which are not drawn to scale, and in which:

(2) FIG. **1** depicts an example architecture of a plurality of data centers, with each data center containing a plurality of servers, with each server communicatively coupled to an external cloud server for managing the server;

(3) FIG. **2** depicts an example of a server with a Remote Access Controller (RAC), illustrating a set of services commonly installed in firmware on a RAC for remotely monitoring and managing the server;

(4) FIG. **3** depicts a system architecture of a plurality of data centers communicatively coupled to an external cloud server, with each data center containing a plurality of servers and a private cloud server, with each server being communicatively coupled to the private cloud server for managing the servers, in accordance with some embodiments;

(5) FIG. **4** depicts a portion of the system architecture of FIG. **3**, illustrating an embodiment of a RAC that may be installed in a server and communicatively coupled to a private cloud server and one or more external cloud servers for supporting the server;

(6) FIG. **5** depicts a portion of the system architecture of FIG. **3**, illustrating one embodiment of a private cloud server with firmware installed and communicatively coupled to a RAC in a server and one or more external cloud servers;

(7) FIG. **6** depicts one embodiment of a deployment architecture for providing available services to a RAC; and

(8) FIG. **7** depicts a flow diagram illustrating one embodiment of a method for discovery and binding of remote access controller firmware to a proxy service.

DETAILED DESCRIPTION

(9) In the following description, details are set forth by way of example to facilitate discussion of the disclosed subject matter. It should be apparent to a person of ordinary skill in the field, however, that the disclosed embodiments are examples and not exhaustive of all possible embodiments.

(10) As used herein, a reference numeral refers to a class or type of entity, and any letter following such reference numeral refers to a specific instance of a particular entity of that class or type. Thus, for example, a hypothetical entity referenced by ‘12A’ may refer to a particular instance of a

particular class/type, and the reference ‘12’ may refer to a collection of instances belonging to that particular class/type or any one instance of that class/type in general.

(11) An information handling system (IHS) may include a hardware resource or an aggregate of hardware resources operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, and/or utilize various forms of information, intelligence, or data for business, scientific, control, entertainment, or other purposes, according to one or more embodiments. For example, an IHS may be a personal computer, a desktop computer system, a laptop computer system, a server computer system, a mobile device, a tablet computing device, a personal digital assistant (PDA), a consumer electronic device, an electronic music player, an electronic camera, an electronic video player, a wireless access point, a network storage device, or another suitable device and may vary in size, shape, performance, functionality, and price. In one or more embodiments, a portable IHS may include or have a form factor of that of or similar to one or more of a laptop, a notebook, a telephone, a tablet, and a PDA, among others. For example, a portable IHS may be readily carried and/or transported by a user (e.g., a person). In one or more embodiments, components of an IHS may include one or more storage devices, one or more communications ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, and a video display, among others. In one or more embodiments, IHS may include one or more buses operable to transmit communication between or among two or more hardware components. In one example, a bus of an IHS may include one or more of a memory bus, a peripheral bus, and a local bus, among others. In another example, a bus of an IHS may include one or more of a Micro Channel Architecture (MCA) bus, an Industry Standard Architecture (ISA) bus, an Enhanced ISA (EISA) bus, a Peripheral Component Interconnect (PCI) bus, HyperTransport (HT) bus, an inter-integrated circuit (I.sup.2C) bus, a serial peripheral interface (SPI) bus, a low pin count (LPC) bus, an enhanced serial peripheral interface (eSPI) bus, a universal serial bus (USB), a system management bus (SMBus), and a Video Electronics Standards Association (VESA) local bus, among others.

(12) In one or more embodiments, an IHS may include firmware that controls and/or communicates with one or more hard drives, network circuitry, one or more memory devices, one or more I/O devices, and/or one or more other peripheral devices. For example, firmware may include software embedded in an IHS component utilized to perform tasks. In one or more embodiments, firmware may be stored in non-volatile memory, such as storage that does not lose stored data upon loss of power. In one example, firmware associated with an IHS component may be stored in non-volatile memory that is accessible to one or more IHS components. In another example, firmware associated with an IHS component may be stored in non-volatile memory that may be dedicated to and includes part of that component. For instance, an embedded controller may include firmware that may be stored via non-volatile memory that may be dedicated to and includes part of the embedded controller.

(13) An IHS may include a processor, a volatile memory medium, non-volatile memory media, an I/O subsystem, and a network interface. Volatile memory medium, non-volatile memory media, I/O subsystem, and network interface may be communicatively coupled to processor. In one or more embodiments, one or more of volatile memory medium, non-volatile memory media, I/O subsystem, and network interface may be communicatively coupled to processor via one or more buses, one or more switches, and/or one or more root complexes, among others. In one example, one or more of a volatile memory medium, non-volatile memory media, an I/O subsystem, a and network interface may be communicatively coupled to the processor via one or more PCI-Express (PCIe) root complexes. In another example, one or more of an I/O subsystem and a network interface may be communicatively coupled to processor via one or more PCIe switches.

(14) In one or more embodiments, the term “memory medium” may mean a “storage device”, a “memory”, a “memory device”, a “tangible computer readable storage medium”, and/or a “computer-readable medium”. For example, computer-readable media may include, without

limitation, storage media such as a direct access storage device (e.g., a hard disk drive, a floppy disk, etc.), a sequential access storage device (e.g., a tape disk drive), a compact disk (CD), a CD-ROM, a digital versatile disc (DVD), a random access memory (RAM), a read-only memory (ROM), a one-time programmable (OTP) memory, an electrically erasable programmable read-only memory (EEPROM), and/or a flash memory, a solid state drive (SSD), or any combination of the foregoing, among others.

(15) In one or more embodiments, one or more protocols may be utilized in transferring data to and/or from a memory medium. For example, the one or more protocols may include one or more of small computer system interface (SCSI), Serial Attached SCSI (SAS) or another transport that operates with the SCSI protocol, advanced technology attachment (ATA), serial ATA (SATA), a USB interface, an Institute of Electrical and Electronics Engineers (IEEE) 1394 interface, a Thunderbolt interface, an advanced technology attachment packet interface (ATAPI), serial storage architecture (SSA), integrated drive electronics (IDE), or any combination thereof, among others.

(16) A volatile memory medium may include volatile storage such as, for example, RAM, DRAM (dynamic RAM), EDO RAM (extended data out RAM), SRAM (static RAM), etc. One or more of non-volatile memory media may include nonvolatile storage such as, for example, a read only memory (ROM), a programmable ROM (PROM), an erasable PROM (EPROM), an electrically erasable PROM, NVRAM (non-volatile RAM), ferroelectric RAM (FRAM), a magnetic medium (e.g., a hard drive, a floppy disk, a magnetic tape, etc.), optical storage (e.g., a CD, a DVD, a BLU-RAY disc, etc.), flash memory, a SSD, etc. In one or more embodiments, a memory medium can include one or more volatile storages and/or one or more nonvolatile storages.

(17) In one or more embodiments, a network interface may be utilized in communicating with one or more networks and/or one or more other information handling systems. In one example, network interface may enable an IHS to communicate via a network utilizing a suitable transmission protocol and/or standard. In a second example, a network interface may be coupled to a wired network. In a third example, a network interface may be coupled to an optical network. In another example, a network interface may be coupled to a wireless network. In one instance, the wireless network may include a cellular telephone network. In a second instance, the wireless network may include a satellite telephone network. In another instance, the wireless network may include a wireless Ethernet network (e.g., a Wi-Fi network, an IEEE 802.11 network, etc.).

(18) In one or more embodiments, a network interface may be communicatively coupled via a network to a network storage resource. For example, the network may be implemented as, or may be a part of, a storage area network (SAN), personal area network (PAN), local area network (LAN), a metropolitan area network (MAN), a wide area network (WAN), a wireless local area network (WLAN), a virtual private network (VPN), an intranet, an Internet or another appropriate architecture or system that facilitates the communication of signals, data and/or messages (generally referred to as data). For instance, the network may transmit data utilizing a desired storage and/or communication protocol, including one or more of Fibre Channel, Frame Relay, Asynchronous Transfer Mode (ATM), Internet protocol (IP), other packet-based protocol, Internet SCSI (iSCSI), or any combination thereof, among others.

(19) In one or more embodiments, a processor may execute processor instructions in implementing at least a portion of one or more systems, at least a portion of one or more flowcharts, at least a portion of one or more methods, and/or at least a portion of one or more processes. In one example, a processor may execute processor instructions from one or more memory media in implementing at least a portion of one or more systems, at least a portion of one or more flowcharts, at least a portion of one or more methods, and/or at least a portion of one or more processes. In another example, a processor may execute processor instructions via a network interface in implementing at least a portion of one or more systems, at least a portion of one or more flowcharts, at least a portion of one or more methods, and/or at least a portion of one or more processes.

(20) In one or more embodiments, a processor may include one or more of a system, a device, and

an apparatus operable to interpret and/or execute program instructions and/or process data, among others, and may include one or more of a microprocessor, a microcontroller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), and another digital or analog circuitry configured to interpret and/or execute program instructions and/or process data, among others. In one example, a processor may interpret and/or execute program instructions and/or process data stored locally (e.g., via memory media and/or another component of an IHS). In another example, a processor may interpret and/or execute program instructions and/or process data stored remotely (e.g., via a network storage resource).

(21) In one or more embodiments, an I/O subsystem may represent a variety of communication interfaces, graphics interfaces, video interfaces, user input interfaces, and/or peripheral interfaces, among others. For example, an I/O subsystem may include one or more of a touch panel and a display adapter, among others. For instance, a touch panel may include circuitry that enables touch functionality in conjunction with a display that is driven by a display adapter.

(22) A non-volatile memory medium may include an operating system (OS) and applications (APPs). In one or more embodiments, one or more of an OS and APPs may include processor instructions executable by a processor. In one example, a processor may execute processor instructions of one or more of OS and APPs via a non-volatile memory medium. In another example, one or more portions of the processor instructions of one or more of an OS and APPS may be transferred to a volatile memory medium and a processor may execute the one or more portions of the processor instructions.

(23) Non-volatile memory medium may include information handling system firmware (IHSFW). In one or more embodiments, IHSFW may include processor instructions executable by a processor. For example, IHSFW may include one or more structures and/or one or more functionalities of and/or compliant with one or more of a basic input/output system (BIOS), an Extensible Firmware Interface (EFI), a Unified Extensible Firmware Interface (UEFI), and an Advanced Configuration and Power Interface (ACPI), among others. In one instance, a processor may execute processor instructions of IHSFW via non-volatile memory medium. In another instance, one or more portions of the processor instructions of IHSFW may be transferred to volatile memory medium, and processor may execute the one or more portions of the processor instructions of IHSFW via volatile memory medium.

(24) Data centers may have large numbers of information handling systems such as servers for processing information. A data center facility may have one or more floors with each floor having racks of servers. A server may be processing a set of information independently or a group of servers may be working on the same set of information.

(25) Turning now to FIG. **1**, an architectural diagram of a plurality of data centers **10** illustrates that each data center **10** may contain a plurality of information handling systems **12** (also referred to as servers **12**). Data center **10-1** may contain M servers **12** and data center **10-2** may contain N servers, where each of M and N may be any number between two to over a thousand. Each server **12** may comprise Remote Access Controller (RAC) **14** to allow remote monitoring and management of any server **12** in data center **10**. RAC **14** may comprise a Dell Remote Access Controller (DRAC) or an integrated Dell Remote Access Controller (IDRAC) for remote monitoring and management.

(26) Turning to FIG. **2**, each server **12** has remote access controller (RAC) **14** with firmware to allow for remote monitoring and managing of server **12**, and server **12** may need to communicate with one or more other servers **12** in data center **10**. Firmware within RAC **14** may include a Representational State Transfer Application Programming Interface (REST API) **16** that allows servers **12** to communicate with each other. Remote monitoring may require communications with processes running in RAC **14**. A D-Bus **18** or other inter-process communications mechanism may enable local communication between processes stored in firmware on RAC **14**.

(27) Firmware in RAC **14** may include a set of built in features **30** executable by a processor in

RAC **14**. For example, each server **12** has a power supply unit (PSU) for receiving power from a power source and transforming the power into usable power by the server **12**. Firmware in RAC **14** includes power control **30-1** as a feature **30** for monitoring of power received by the PSU to determine when server **12** has switched power modes and allow for remote switching of server **12** between power modes. Similarly, server **12** has a fan for cooling server **12** and firmware in RAC **14** includes thermal control **30-2** as a feature **30** for monitoring of temperatures in server **12** to determine if the fan is operating, determining when to operate the fan and enabling remote operation of the fan.

(28) Firmware in RAC **14** may also store hardware abstraction layer (HAL) **22** for communication with and monitoring of peripheral devices. Within HAL **22**, features such as Inter-Integrated Circuit (I2C) protocol **22-1**, Improved Inter-Integrated Circuit (**13**C) protocol **22-2**, Serial Peripheral Interface (SPI) **22-3**, Reduced Media-Independent Interface (RMII) **22-4**, Peripheral Connect Interface Express Vendor Defined Message (PCIe VDM) **22-5** and management component transport protocol (MCTP) **22-6** allow communications with chips, processors and microcontrollers in server **12**.

(29) Server **12** includes hardware **24** such as processors for processing information, memory for storing information, a fan for cooling server **12** including devices and components in server **12**, a network integrated circuit (NIC) for communications, other controllers such as RAID controllers, and Complex Logic Programmable Devices (CPLDs). Accordingly, Firmware in RAC **14** may include other features such as Basic Input Output Service (BIOS) **26-1**, NIC **26-2**, fan control **26-3**, PSU control **26-4** for operating a PSU, RAID feature **26-5** for managing a RAID and CPLD feature **26-6** for monitoring and managing a CPLD.

(30) Firmware in RAC **14** may typically include other features **30** for monitoring and managing servers **12** in data centers **10**. The examples depicted in FIG. **3** and provided below represent more common features found in RAC **14**, and RAC **14** may store additional features **30** not listed.

(31) Web Graphical User Interface (GUI) **30-1** is a web-based application that processes network events for displaying the data in a graphical format for users to view on an attached display. Redfish® 30-2 is an application programming interface (API) that uses RESTful semantics to access data defined in model format for systems management. Remote Access Controller admin (RACADM) feature **30-3** is a scriptable interface to allow remote configuration of RAC **14**. Simple Network Management Protocol (SNMP) feature **30-4** may be used to collect data related to network changes and determine the status of network-connected devices. Hardware inventory feature **30-5** may maintain an inventory and properties of all hardware installed on server **12**. Software inventory feature **30-6** may maintain an inventory and versions of all software running on server **12**. Firmware update feature **30-7** may maintain a list of firmware including versions and facilitate updates of any firmware on server **12**. Monitoring feature **30-8** may monitor operation of components or devices in server **12** and record values or may define what operations are to be monitored and how the monitoring should occur. RAC **14** may include database **32** for storing information about components or devices in server **12**. Network Controller—Sideband Interface (e.g., NC-SI 1.2) feature **30-10** defines a control communication protocol between a baseboard management controller (BMC) and one or more Network Interface Controllers (NICs). Platform Level Data Model (PLDM) feature **30-11** defines the contents of a firmware update package. Security Protocol and Data Models (SPDM) feature **30-12** enables authentication, attestation and key exchange for enabling security of server **12**. Management Control Transport Protocol (MCTP) feature **30-13** stores message formats, transport descriptions, message exchange patterns and endpoint characteristics related to communications between components.

(32) Servers **12** described above have several shortcomings. All features **30** are stored in memory in each RAC **14** of each server **12**, regardless of whether a feature **30** will be used for a particular server **12**. Furthermore, features **30** are getting more robust and require more memory to manage devices in servers **12**. For example, RAC **14** in some generations of servers **12** may have 512

Megabytes (MB) of memory to store all features **30**, RAC **14** in later generations of servers **12** may have 1 Gigabyte (GB) of memory to store all features **30**, and RAC **14** in later generations of servers **12** may have 2 GB of memory to store all features **30**. Although each server **12** is provided with the full suite of features **30**, many customers or servers **12** require a limited set of features **30**. For security reasons, some users prefer to have only firmware code which they know they will use. Subscription-based licensing, in which a user pays for only the features needed, may be preferred. However, only about 35% of datacenter servers **12** contain the latest firmware for RAC **14**, and features **30** are often unnecessarily tied to specific hardware capabilities. Customization of features **30** via Software Developer Kits (SDK) may be limited or unavailable due to hardware capabilities or software requirements. Some devices (e.g., Baseboard Management Controllers (BMCs)) are expected to be commoditized, resulting in greater complexity and/or more features **30** that would need to be installed on each RAC **14**.

(33) Turning to FIGS. **3**, **4** and **5**, embodiments disclosed herein allow a remote access controller (RAC) **314** in server **312** to have firmware that has a minimum number of features **30** that can be augmented as needed with any number of available services **350** executing on a private cloud server **360**. In some embodiments, RAC **314** may be manufactured with firmware including a minimum Hardware Abstraction Layer (HAL). Some controls **20** and features **30** may remain as built-in controls **20** or features **30** in firmware in RAC **314**. For example, any time server **12** is operating, PSU control **20-1** monitors power supplied to the PSU and is therefore maintained in firmware in RAC **314**. Embodiments may augment built-in features **30** with available services **350** that can be retrieved from external cloud server **210** in cloud **4** and installed and executed on private cloud server **360** or possibly installed on RAC **314**, discussed in greater detail below. If needed by RAC **314**, available services **350** may be delivered/deployed by leveraging and extending on top of existing firmware in RAC **314**.

(34) Advantageously, embodiments may provide vendor-agnostic, unified and seamless system management across generations of servers **312**. Embodiments may further provide accelerated delivery of new available services **350** with subscription-based granular licensing.

(35) Referring to FIG. **3**, data centers **310** may each have multiple servers **312**. Data center **310-1** may have a first plurality of servers **312** (e.g., 2 to M) divided into P groups **340**, wherein group **340-1** may have a first set of servers **312** (e.g., 1 to A), group **340-2** may have a second set of servers **312** (e.g., 1 to B) and group **340-**P may have a third set of servers **312** (e.g., 1 to C). Data center **310-1** may further comprise private cloud server **360-1** communicatively coupled to all servers **312-1** to **312-**M in data center **310-1**. Private cloud server **360-1** may be communicatively coupled to external cloud server **210** in cloud **4**.

(36) Data center **310-2** may have a second plurality of servers **312** (e.g., **2** to N) divided into Q groups **340**, wherein group **340-4** may have a first set of servers **312** (e.g., **1** to D), group **340-2** may have a second set of servers **312** (e.g., **1** to E) and group **340-**Q may have a third set of servers **312** (e.g., **1** to F). Data center **310-2** may further comprise private cloud server **360-2** communicatively coupled to all servers **312-1** to **312-**N in data center **310-2**. Private cloud server **360-2** may be communicatively coupled to external cloud servers **210** in cloud **4**.

(37) Groups **340** may be logical groups, discussed in greater detail with respect to FIG. **6**.

(38) Referring to FIG. **4**, external cloud server **210** may communicate with private cloud servers **360** over a network to provide available services **350**.

(39) Cloud integration service **210-1** may include data monitoring **214** and cloud services **216**. Dell CloudIQ is an example of a data monitoring service **214** that may monitor operation of servers **312** in data centers **310** for data protection. Dell APEX is an example catalog of cloud services **216** that may integrate infrastructure and software for cloud operations.

(40) Infrastructure Solutions Group (ISG) server **210-2** may provide storage and networking services, and may include third-party services **350-18**. In some embodiments, ISG server **210-2** may provide RAC proxy on Data Processing Unit (DPU) service **220** to private cloud server **360** to

offload processing.

(41) Management Service Catalog (MSC) server **210-3** may store a database of available services **350** that can be provided to private cloud servers **360**. In some embodiments, private cloud server **360** communicates with MSC server **210-3** to retrieve available services **350**. In some embodiments, MSC server **210-3** communicates with private cloud server **360** to install available services **350**.

(42) IOT integration service **210-4** may enable devices to communicate with other devices and enable external cloud servers **210** to communicate with or monitor devices in servers **312**.

(43) Autonomous server manager **210-5** may provide server management without a user.

(44) Turning to one or more of FIGS. **4** and **5**, servers **312** may be similar to servers **12** depicted in FIG. **2** in that they comprise hardware such as a PSU for receiving power from a source and transforming the power into usable power by the server, a fan, network IC (NIC) for communications, other controllers such as RAID controllers, and Complex Logic Programmable Devices (CPLDs). Servers **312** also include BIOS **26-1**, NIC **26-2**, fan control **26-3**, PSU service **26-4** for operating a PSU, remote controller service **26-5** for managing a RAID and CPLD service **26-6** for monitoring and managing CPLDs.

(45) Firmware within RAC **314** on servers **312** may include a Representational State Transfer Application Programming Interface (REST API) **16** that allows servers **312** to communicate with each other. Remote monitoring may require communications with processes running in RAC **314**. A D-Bus **18** or other inter-process communications mechanism enables local communication between processes on RAC **314**. RAC **314** may further comprise Remote Procedure Call (RPC) queue **302**, D-Bus to RPC mapper **304** and D-Bus to REST mapper **306** for available services **350** on private cloud server **360** to communicate with HAL **22**. In some embodiments, RPC mapper **304** may be a Google RPC (gRPC) mapper **304**.

(46) RAC **314** may also store hardware abstraction layer (HAL) **22** for communication and monitoring of peripheral devices. Within HAL **22**, I2C **22-1**, **13**C **22-2**, SPI **22-3**, RMII **22-4**, PCIe VDM **22-5** and MCTP **22-6** allow communications with chips, processors and microcontrollers in server **312**.

(47) Available Services May be Stored and Executed on a Private Cloud Server

(48) Embodiments of private cloud server **360** may store a set of available services **350** retrieved from MSC server **210-3**. In some embodiments, private cloud server **360** may determine, based on one or more of an application to be executed on a server **312** or a data set to be processed by a server **312**, that a particular available service is needed and communicate with MSC server **210-3** to retrieve the available service **350**. In some embodiments, private cloud server **360** may communicate with MSC server **210-3** to retrieve an available service **350** based on a subscription. Available services **350** may be executed by private cloud server **360** unless there is a need to have RAC **314** execute the available service **350**.

(49) Available Services May be Installed in a Rac if Needed

(50) Embodiments may install available services **350** in RAC **314** from private cloud server **360**, wherein the set of available services **350** stored in RAC **314** may be less than the total number of available services **350** stored in private cloud server **360** and less than the plurality of available services **350** stored in MSC server **210-3**. Available services **350** may be installed in RAC **314** based on one or more of performance or capability. Once the process is complete or the available service **350** is no longer time-sensitive, embodiments may uninstall the available service **350** from RAC **314**.

(51) As a device-based example, referring to available service **350-13** (i.e., MCTP **350-13**) and available service **350-10** (i.e., NC-SI 1.2 protocols **350-10**), if a process sends a request to CPLD **26-6**, a response may be expected in a minimum response time of about 50 milliseconds (ms) and a maximum response time of about 4 seconds. If the MCTP **350-13** and NC-SI 1.2 protocols **350-10** are stored in private cloud server **360**, a response may take 5 seconds. Embodiments may install

NC-SI 1.2 protocols **350-10** as a time sensitive plug-in **340** to ensure a response is received in less than 4 seconds. Once NC-SI 1.2 protocols **350-10** is not needed, NC-SI 1.2 protocols **350-10** may be uninstalled from RAC **314**.

(52) As another example, referring to available service **350-11** (i.e., PLDM **350-11**) and available service **350-7** (i.e., Firmware update service **350-7**), if there is a firmware update and PLDM **350-11** executing on private cloud server **360** cannot deliver the update payload within a maximum time (e.g., 4 seconds), the firmware update may time out. In this case, PLDM **350-11** and Firmware update service **350-7** may be installed as time sensitive plug-ins **340** to ensure the firmware update payload can be delivered in time. Once the firmware update payload is delivered, PLDM **350-11** and Firmware update service **350-7** may be uninstalled from RAC **314**.

(53) Existing Available Services Stored on Private Cloud Server

(54) Box **352** contains a non-limiting list of available services **350** that may be installed on private cloud server **360** to communicate with RACs **314** on any servers **312** in data centers **310**, wherein available services **350** in box **352** may function similar to features **30** described above with respect to FIG. **2** but execute on private cloud server **360**. For example, RAC **314** may be configured with a minimum number of features **30** and private cloud server **360** may store and execute Web GUI service **350-1**; Redfish® server management service **350-2**; RACADM command-line service **250-3**; SNMP service **350-4**; Hardware (HW) inventory service **350-5** for tracking information and properties of hardware installed on servers **312**; Firmware (FW) inventory service **350-6** for tracking information and versions of firmware installed on servers **312**; FW update service **350-7** for installing Firmware updates in servers **312**; Monitoring service **350-8** for coordinating monitoring of operation of servers **312**; Unified database **350-9** for storing information for sharing among processes and applications executing on servers **312**; NC-SI service **350-10**; PLDM service **350-11**; SPDM service **350-12** and MCTP service **350-13**. In some embodiments, available services **350** may be retrieved from Management Service Catalog server **210-3** through subscriptions, wherein any available service **350** may be retrieved as needed.

(55) New Services and Services with Increased Scope

(56) Embodiments may allow private cloud server **360** to provide additional available services **350** and available services **350** with increased scope that increase the capabilities of RAC **314**. Box **354** contains a non-limiting list of available services **350** that may be retrieved from MSC server **210-3** and installed on private cloud server **360**. Some available services **350** in box **354** may be executed by private cloud server **360**. Some available services **350** in box **354** may be installed on RAC **314** on any server **312** in data center **310**.

(57) Credentials Service

(58) A user must have administrative credentials to run RACADM commands remotely. When a user wants to run RACADM commands, the user must first be validated by RAC **314**. Traditionally, RAC **14** corresponding to server **12** in FIG. **2** may store credentials for one to about sixteen users. Embodiments described herein allow private cloud server **360** to download and execute credentials service **350-15**, which may be configured to manage credentials for one user to more than sixteen users and map the credentials to RACs **314** as needed. In some embodiments, unified database **350-9** on private cloud server **360** may be configured to store credentials for a large number of users (e.g., over a thousand) to increase the scope of credentials service **350-9** over credentialing features **30** commonly found in RACs **14**. In some embodiments, private cloud server **360** may be configured to install credential service **350-15** and store a database in unified database **350-9** with the credentials for any number of users between one to over a thousand. As the number of users changes, private cloud server **360** may update the database. In some embodiments, credentials service **350-15** may validate users. In some embodiments, credentials service **350-15** may register itself as a credentials manager, receive requests, and validate users such that RAC **314** does not validate but knows the user is validated.

(59) Telemetry Service

(60) Some Baseboard Management Controllers (BMCs) have no intelligence to adjust telemetry collection based on server events/errors or have different streaming rate for each metric. Traditionally, RAC **14** in server **12** may send the same information multiple times or at different rates, tying up networking resources and memory. Telemetry service **350-16** may be retrieved from MSC server **210-3** and installed on private cloud server **360**. In some embodiments, an Open Telemetry (OTEL) service **350-16** may be stored in private cloud server **360** as an available service **350**. When servers **312** communicate data for telemetry purposes, telemetry service **350-16** executing on private cloud server **360** may aggregate data, remove redundant data, or otherwise coordinate the communication of data, resulting in reduced network congestion. In some embodiments, telemetry service **350-16** may be installed in RAC **314** to meet telemetry requirements and then uninstalled after server **312** does not need to meet any telemetry requirements. In some embodiments, telemetry service **350-16** may be installed in RAC **314** as a time-sensitive plug-in **340** to provide quicker responses to telemetry requirements and then uninstalled after server **312** does not need quick responses to meet telemetry requirements.

(61) AI/ML Service

(62) Artificial Intelligence (AI)/Machine learning (ML) service **350-17** may include services necessary for AI/ML. If a server **312** (or a set of servers **312**) is needed for AI/ML, AI/ML service **350-17** may be downloaded to private cloud server **360** for coordinating processing by servers **312** for AI/ML processing.

(63) Third Party Services

(64) Third party services **350-18** may include services needed for particular third-party applications. Advantageously, instead of all available services **350** being tied to particular hardware, embodiments may enable third-party services **350-18** to execute on private cloud server **360**, wherein other available services **350**, RPC queue **302**, D-Bus to RPC mapper **304** and D-Bus to REST mapper **306** enable third-party services **350-18** on private cloud server **360** to communicate with HAL **22** in one or more RACs **314**.

(65) Advanced Power/Thermal Service

(66) Advanced power/thermal service **350-19** may be stored on private cloud server **360** and may refer to an available service **350** that can be executed on private cloud server **360** to communicate with RACs **314** on multiple servers **312** to monitor or control power or thermal operations of one or more servers **312**. For example, data center **310** may have multiple floors with hundreds of racks of servers **312**. Each RAC **314** may communicate with sensors inside a server **312** for remotely and individually monitoring temperature of that server **312**. Advanced power/thermal service **350-19** may allow a user to remotely and collectively monitor and manage temperatures for multiple servers **312**, such as all servers **312** processing a set of information, all servers **312** in a rack or all servers **312** on a floor. In some embodiments, advanced power/thermal service **350-19** may be installed in RAC **314** (e.g., as a time-sensitive plug-in **340**) for quicker response to power/thermal requirements and then uninstalled after server **312** does not need to operate under advanced power/thermal requirements.

(67) Secure Zero Touch Provisioning (sZTP) Service

(68) Secure Zero-Touch Provisioning (sZTP) service **350-20** enables a server **312** to securely provision a network device when booting in a default state. If server **312** is expected to require communication over a network, sZTP service **350-20** executing on private cloud server **360** may ensure the network device is securely provisioned. In some embodiments, sZTP service **350-20** may be installed in RAC **314** for booting and uninstalled once server **312** has successfully booted. In some embodiments, sZTP service **350-20** may be installed on private cloud server **360** and installed on servers **312** (e.g., as a time-sensitive plug-in **340**) as needed to ensure network devices are securely provisioned.

(69) ISM Services

(70) Integrated Dell Remote Access Controller (iDRAC) Service Manager service **350-21** may

refer to available services **350-21** that may be executed to monitor and manage operation of RAC **314**.
(71) Fido Device Onboard (FDO) Service
(72) Fast IDentity Online (FIDO) Device Onboard (FDO) service **350-22** allows onboarding and trusting of new devices (e.g., RAC **314**) within an environment. For example, RAC **314** may be Linux-based and one or more available services **350** may be Windows-based. FDO service **350-22** may enable servers **312** to execute Linux-based services **350** and Windows-based services **350**. Data centers **310** may have hundreds or thousands of servers **312**. Devices in servers **312** may be removed and exchanged for newer devices. Instead of a remote user configuring each device and deploying required applications, FDO service **350-22** allows multiple devices in various servers **312** to be configured correctly. In some embodiments, FDO service **350-22** may be installed in RAC **314** for quicker configuring and deployment and uninstalled once server **312** has successfully configured and deployed new devices. In some embodiments, FDO service **350-22** may be installed on private cloud server **360** and installed on servers **312** (e.g., as a time-sensitive plug-in **340**) as needed to ensure new devices are quickly configured and deployed.
(73) Internet of Things (IOT) Integration Service
(74) IOT Integration service **350-23** may be installed on private cloud server **360** to facilitate integration between devices on servers **312**. Advantageously, instead of each server **312** in multiple data centers **310** communicating with IOT integration server **210-4**, private cloud server **360** may perform some of the integration, wherein other available services **350**, RPC queue **302**, D-Bus to RPC mapper **304** and D-Bus to REST mapper **306** enable private cloud server **360** to communicate with HAL **22** in one or more RACs **314** in servers **312** that may have different devices.
(75) Feature Tour
(76) Feature Tour service **350-24** may be installed to guide users through features **30** available to server **312**. Advantageously, instead of each RAC **314** in each server **312** in multiple data centers **310** storing all the information necessary to guide users through only features **30** installed on that server, private cloud server **360** may store all the information and provide information including features **30** and available services **350** available to a particular server **312**. An end user may not know whether a feature is installed on server **312** or is instead an available service **350** accessible by server **312**. In some embodiments, private cloud server **360** may install feature tour service **350-24** with information relevant to a particular server **312** on RAC **314** associated with the particular server **312**.
(77) Infrastructure Programmer Developer Kit (IPDK) Service
(78) IPDK service **350-25** is an open-source, vendor-agnostic framework of drivers and APIs. IPDK service **350-25** may be too large to install on each RAC **314** in each server **312**. IPDK service **350-25** may be installed on private cloud server **360** for infrastructure offload and management.
(79) For each available service **350**, if the available service **350** is needed for a particular server **312**, a version of the available service **350** may be retrieved from external cloud server **210** and executed on private cloud server **360**, wherein RAC **314** contains a minimum number of built-in controls **20** and features **30** to communicate with private cloud server **360**, wherein private cloud server **360** comprises memory and processors for executing the available service **350**. If, at some later time, the available service **350** is not needed on a particular server **312**, the available service **350** may be uninstalled from RAC **314** but a version may still be stored in private cloud server **360** or the available service **350** may be uninstalled from private cloud server **360**.
(80) MSO—Management Service Orchestrator
(81) MSO **356** may coordinate between features and ensure any service **350** is compatible with other available services **350** and hardware in server **312** and that no conflicts exist. For example, regarding telemetry, a manufacturer may have a default telemetry service **20** installed in firmware, but a customer may want to use Open Telemetry (OTel) service **350-16** or a third-party telemetry

service **350-18**. If the customer requests another telemetry service, MSO **356** may determine whether the requested telemetry application will work, if an existing telemetry feature needs to be disabled or an existing telemetry service **350-16** or **350-18** needs to be uninstalled.
(82) Large Send Offload (LSO)
(83) Large Send Offload (LSO) service **358** may increase the egress throughput of high-bandwidth network connections. LSO service **358** may be installed on PCS **312**, reducing workload performed by RAC **314**. In some embodiments, LSO **358** enables communication with RAC Proxy on DPU service **220** to offload large workloads.
(84) Secure Discovery and Binding of Rac Firmware to a Rac Proxy Service
(85) Referring to FIG. **6**, embodiments may securely discover and bind RAC firmware to a RAC proxy service in private cloud server **360**.
(86) On server **312**, a first RAC firmware **316-1** may store a provisioning client service **602**A, a Credentialing Authority (CA) certificate **604**A and a RAC hardware identity certificate **606**A and a second (redundant) RAC firmware **316-2** may store a provisioning client service **602**B, a Credentialing Authority (CA) certificate **604**B and a RAC hardware identity certificate **606**B. In some embodiments, RAC hardware identity certificates **606** may be stored in RAC firmware **316** during manufacturing of server **312**.
(87) Private cloud server **360** may store dynamic host configuration protocol (DHCP) service **610**, onboarding service **612**, RAC proxy service **614**, signed CA certificate **616** and secure credential vault **618**.
(88) Public cloud server **210** may comprise manufacturer authorized signing authority (MASA) service **620**.
(89) Validating a chain of trust may provide mutual authentication of the RAC firmware **316** and RAC proxy service **612**. As depicted in FIG. **6**, in cases where server **312** comprises multiple RAC firmwares **316** (e.g., a dual firmware configuration), RAC proxy service **612** may identify redundant RAC firmware (e.g., RAC firmware **316-2**) and associate all the bindings between RAC firmwares **316** and RAC proxy service **612**.
(90) Secure Binding
(91) In some embodiments, a credentialing authority (CA) signs a hardware identity certificate **606** and embeds a CA certificate **604** into RAC firmware **316** during a manufacturing process. Also during a manufacturing process, the CA may sign a proxy service certificate **614** and embed the proxy service certificate **614** in private cloud server **360**. A private key may be stored in secure credentialing vault **616**.
(92) When RAC **314** initiates binding with onboarding binding service **612**, RAC **314** may communicate a signed binding request with a RAC hardware root of trust. Onboarding binding service **612** may validate the binding request by determining if the RAC hardware root of trust is signed by the CA.
(93) If onboarding binding service **612** determines the binding request is valid, onboarding binding service **612** may bind RAC firmware **316** with an onboarding certificate and send a response to RAC **314**.
(94) When the response is received by RAC **314**, RAC **314** may determine whether an onboarding certificate **614** is signed by the credentialing authority (CA). If so, RAC firmware **316** is bound to onboarding binding service **612**.
(95) FIG. **7** depicts a flow diagram **700** of a method for secure discovery and binding of RAC firmware to a RAC proxy service.
(96) At step **702**, when RAC **314** is powered on, RAC **314** executing instructions in RAC firmware **316** communicates with DHCP service **610** and gets an RAC Internet Protocol (IP) address.
(97) At step **704**, private cloud server **360** provides the details about the onboarding service **612**.
(98) At step **706**, RAC **314** communicates with onboarding service **612** via a priority provisioning protocol.

(99) At step **708**, RAC **314** validates the identify of onboarding service **612**.
(100) At step **710**, RAC **314** provides a RAC hardware identity certificate **606**. In some embodiments, RAC hardware identity certificate **606** comprises a SUDI Cert (Secure Unique Device Identity Certificate).
(101) At step **712**, onboarding service **612** validates RAC hardware identity certificate **606**.
(102) At step **714**, embodiments determine if there is mutual authentication. If not, then at step **716**, the process ends. If there is mutual authentication, onboarding service **612** generates a token associated to RAC hardware identity certificate **606** at step **718**.
(103) At step **720**, onboarding service **612** provides the RAC proxy service details and the token to RAC firmware **316**. In some embodiments, the token comprises an Open Authentication (OAUTH) token.
(104) At step **722**, using the token, RAC **314** communicates with a RAC proxy service **614**.
(105) At step **724**, RAC proxy service **64** validates the token and challenges RAC **314** to provide RAC hardware identity certificate **606**.
(106) At step **726**, embodiments check for authentication of RAC hardware identity certificate **606**. If there is no authentication, then the process ends at step **716**.
(107) At step **728**, if embodiments authenticate RAC hardware identity certificate **606**, RAC proxy service **614** provides CA certificate **616**. In some embodiments, RAC proxy service **614** stores a private key in Secure Credential Vault **618**.
(108) At step **730**, RAC **314** validates the root of trust of the RAC proxy service certificate **616** and challenges the RAC proxy service **614** to prove ownership of CA certificate **616**.
(109) At step **732**, embodiments may attempt to authenticate CA certificate **616**. If CA certificate **616** is not authenticated, the process end at step **716**.
(110) If embodiments authenticate CA certificate **616**, at step **734**, RAC proxy service **614** establishes a binding to RAC firmware **316**.
(111) At step **736**, RAC proxy service **614** may check for a response from RAC **314**. If there is a response, at step **738**, RAC proxy service **614** may continue binding to RAC firmware **316**.
(112) At step **740**, if there is no response from RAC **314**, RAC proxy service **614** may check a heart-beat mechanism for RAC **314**.
(113) If RAC proxy service **614** determines there is a heart-beat, at step **738**, RAC proxy service **614** may continue binding to RAC firmware **316**.
(114) If RAC proxy service **614** determines there is not a heart-beat, at step **742**, RAC proxy service **614** may remove the binding and the process ends at step **716**.
(115) When RAC firmware **316** is bound to RAC proxy service **614**, embodiments may securely install available services **350** in RAC firmware **316** and communicate telemetry data and other information.
(116) The above disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other embodiments which fall within the true spirit and scope of the present disclosure. Thus, to the maximum extent allowed by law, the scope of the present disclosure is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

Claims

1. A method, comprising: communicating, by a remote access controller (RAC) on a server, with a private cloud server to get a RAC Internet Protocol (IP) address from a Dynamic Host Configuration Protocol (DHCP) service; determining information associated with an onboarding service; communicating, based on the IP address and the DHCP service, with the onboarding service to validate the identity of the onboarding service; providing a RAC hardware identity

certificate; validating, by the onboarding service, the RAC hardware identity certificate; generating, by the onboarding service, a token associated with the RAC hardware identity certificate; sending, by the onboarding service, the token and information about a RAC proxy service to the RAC; validating, by the RAC proxy service, the token; providing, by the RAC, the RAC hardware identity certificate; sending, by the RAC proxy service, a RAC proxy service certificate to the RAC; validating, by the RAC, a root of trust of the RAC proxy service certificate; and establishing, by the RAC proxy service, a binding to RAC firmware in response to successful validation of the root of trust of RAC proxy service certificate.

2. The method of claim 1, wherein the RAC hardware identity certificate comprises a Secure Unique Device Identity (SUDI) Certificate.

3. The method of claim 1, wherein the token comprises an open authentication (OAUTH) token.

4. The method of claim 1, further comprising the RAC proxy service using a heart-beat mechanism to keep the binding active.

5. The method of claim 4, wherein if there is no response from the RAC, removing the binding.

6. The method of claim 1, wherein determining information associated with the onboarding service comprises receiving, from a provisioning service, information associated with the onboarding service.

7. The method of claim 1, wherein the RAC proxy service certificate comprises a central certificate authority (CA) certificate.

8. The method of claim 1, wherein validating a root of trust of the RAC proxy service certificate comprises determining if the CA certificate is signed.

9. A data center comprising: a plurality of servers, wherein each server comprises: a remote access controller (RAC) comprising: a RAC processor; and a RAC firmware storing: a central certificate authority (CA) certificate; and a RAC hardware identity certificate; wherein at least one server of the plurality of servers comprises a private cloud server (PCS) communicatively coupled to the plurality of servers and an external cloud server, the private cloud server comprising: a PCS processor; and a PCS memory storing: a dynamic host configuration protocol service configured to communicate an Internet Protocol (IP) address to the RAC; an onboarding service configured to: validate the RAC hardware identity certificate; generate a token associated with the RAC hardware identity certificate; and send the token and information about a RAC proxy service to the RAC; and a RAC proxy service configured to: validate the token; and send a RAC proxy service certificate to the RAC; wherein the RAC is configured to validate a root of trust of the RAC proxy service certificate; and the RAC proxy service is configured to establish a binding to the RAC firmware in response to successful validation of the root of trust of RAC proxy service certificate.

10. The data center of claim 9, wherein the RAC hardware identity certificate comprises a Secure Unique Device Identity (SUDI) Certificate.

11. The data center of claim 9, wherein the token comprises an open authentication (OAUTH) token.

12. The data center of claim 9, wherein the RAC proxy service is configured to use a heart-beat mechanism to keep the binding active.

13. The data center of claim 12, wherein if there is no response from the RAC, the RAC proxy service is configured to remove the binding.

14. The data center of claim 9, further comprising a provisioning service configured to provide information associated with the onboarding service.

15. The data center of claim 9, wherein the RAC proxy service certificate comprises a central certificate authority (CA) certificate.

16. The data center of claim 15, wherein to validate the root of trust of the RAC proxy service certificate, the RAC is configured to determine if the CA certificate is signed.